2 Italy

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The main objective of this chapter is to construct a reliable general chronology of Italian famines, building upon earlier attempts but also taking into account the results of recent research, as well as conducting systematic new tests on previously unavailable databases. Such tests allow us both to confirm widespread ideas about the significance of some specific events and to challenge the received wisdom about some others (especially for the eighteenth century). Additionally, we are attempting for the first time to integrate studies of famines in specific epochs into a very long-run picture: approximately from 1250 up until 1810. Although tracing and measuring the occurrence of famines over time is our main aim, we also provide some information about their demographic and socio-economic consequences and about the role played by institutions charged with managing food provisioning. We conclude with a synthetic discussion of causality, developing an analysis of a whole range of factors potentially involved in originating a famine, including population dynamics, climate change, war and pestilence, institutional failures, the conditions of food production and distribution and changes in crops and in agrarian technologies.

1 Reconstructing a Chronology of Italian Famines

Two partial chronologies of famines affecting Italy in the late medieval and early modern period exist. They rely on different methods for identifying the main events. The older chronology is Del Panta and Livi Bacci's classic reconstruction of Italian mortality crises, 1600–1850, which also

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includes a (partial) classification of the crises distinguishing epidemics (for example, of plague or typhus), famines and others. A more recent attempt is Alfani's reconstruction of famines in northern Italy, 1470–1700, which uses an 'expert' method – i.e. one reliant upon a number of case studies produced by scholars specialised in the history of famines or food provisioning to identify the main events. One of the explicit objectives of this method is 'to solve, at least in part, the problem of differentiating "famine" from simple "dearth"' (Alfani 2015).

These pre-existing chronologies are summarised and compared in Table 2.1. Our aim is to produce a general chronology of Italian famines that integrates, updates and verifies them and, additionally, expands them in time (covering the whole period from about 1250 until today) and space (covering the whole of the Italian Peninsula). To this end, we will employ a variety of methods, to make full use of the information and of the sources available for different epochs and areas.

Ó Gráda's definition of famines offers a clear identification of what should be considered the main marker for the occurrence of a 'famine': excess mortality – i.e. an increase in mortality over the 'normal' level, caused by either starvation or hunger-induced diseases, such as typhus. A simple dearth might cause hunger but does not kill people, while a proper famine is a killing event – and, as such, it requires an acute and prolonged period of hunger, as the human body can resist being seriously deprived of food for very long periods (Livi Bacci 2000), or at least it has to be debilitating enough for a sufficiently large part of the population to favour the spread of disease. The link between famine and disease is a particularly complex one and has been developed in detail elsewhere (Alfani 2013a: 43–6), including with specific reference to the Italian case. Here it will suffice to note that, to begin with, what we are interested in is the outcome of famine in terms of deaths, and not so much how those deaths were 'produced', which should be considered simply as part of the description of specific events, or as an element to explore in order to provide causal explanations for the occurrence of such events (some examples will be provided below).

Del Panta and Livi Bacci's method, then, which focuses on mortality trends to identify periods of crisis, seems to be very well suited to our aims and offers a number of advantages. First of all, to be applied it requires only time series of deaths (burials): a kind of information that is fairly abundant across Europe from the early modern period, as parish records of burials exist quite systematically from the early seventeenth century. The Roman Catholic Church made them mandatory for all parishes with the *Rituale Romanum*, introduced in 1614 (Alfani, Dalla Zuanna and Rosina 2009), and similar sources exist for most Protestant

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Table 2.1 *A comparison of the main famine chronologies available in the historical literature, 1470–1810*

Del Panta and Livi Bacci	Alfani
	1472–1474
	1476
	1482–1484
	1493
	1502–1505
	1518–1520
	1527–1529
	1533–1534
	1539–1540
	1544–1545
	1550–1552
	1558–1560
	1569–1572
	1586–1587
	1590–1593
	1600–1601
1607–1608	1608
1622	1618–1622
1629	1628–1632
1648–1649*	1648–1649
1672	
1678–1679	1678–1679
	1693–1695
1709	
1716	
1724	
1764	
1767	
1801	

Notes: * Del Panta and Livi Bacci attribute this crisis to typhus (see discussion in the following).
Sources: Elaboration from Del Panta and Livi Bacci (1977) and from Alfani (2015). The first part of Alfani's chronology, covering 1470–1627, was first published (including greater detail, as well as a comparison with plague chronology) in Alfani (2013a: 176–7).

areas from about the same period. However, parish books of burials are rare before 1600 and almost non-existent before 1550, even in an area particularly rich in this kind of documentation such as Italy. Before 1600 other sources exist, the city books of the dead, that have characteristics similar to the parish books of burials and are sometimes available since

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the fifteenth century. However, these can be found for the main urban settlements only, and, consequently, any method relying upon time series of deaths or burials could not be applied systematically before the middle of the sixteenth century.

Del Panta and Livi Bacci define a mortality crisis as a short-term perturbation of mortality that reduces the dimension of the generations so much that they are unable to reproduce themselves entirely even making full use of their potential for recovery. A mortality crisis, then, happens when one generation is prevented from giving birth to another at least equal in size, even when the rise in fertility and nuptiality that always follows a peak of deaths is taken into account. A 50 per cent rise in deaths is enough to prevent the generation born in the year of the crisis from fully reproducing. This would be a 'small' crisis. A 300 per cent rise in deaths could not be counterbalanced by the recovery potential of all the generations under the age of 15 at the moment of the crisis. This would be a 'major' crisis. Key to the application of this process is the measurement of 'normal' mortality – i.e. the number of deaths compared to which a peak level can be defined as a crisis (if it exceeds by 50 per cent or more than 'normal') or not. Del Panta and Livi Bacci suggest calculating the normal level of deaths by recurring to an 11-term moving average, centred on the year under consideration and excluding the two higher and the two lower terms (Del Panta and Livi Bacci 1977: 409–10). While this method is indeed able to identify short-term perturbations in mortality levels that might have to be ascribed to a real crisis, its practical application presents some difficulties, especially in the presence of time series with frequent gaps (as is usually the case with the older time periods, especially in rural communities) or in the aftermath of severe mortality crises able to alter for many years the normal death levels (such as the 1630 plague, which in northern Italy killed about one-third of the overall population).

Consequently, a variant of the original method, as introduced by Alfani in his study of the 1630 plague (Alfani 2013b), has been used. The difference lies in the definition of the normal level of deaths, which is taken as the five-term average, maximum and minimum values excluded, covering from the sixth to the second year preceding the first crisis year (the year immediately preceding the crisis is omitted, to avoid pre-peak disturbances in the deaths level). Crisis mortality is taken as the maximum yearly mortality to be found in the whole crisis period, and is then compared to the normal level. The method is particularly well suited for identifying the communities affected by a *known* possible crisis – which is exactly our case, as we can make use of pre-existing chronologies to

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test the extent of given famine-induced crises. The interpretation of the outcome, in terms of the ability of generations to fully reproduce, does not change compared to the original model.

As can easily be seen from Table 2.1, for the period during which they overlap (the seventeenth century) the two earlier chronologies match very well – especially if one considers that the 1672 famine reported by Del Panta and Livi Bacci affected, according to their estimates, only southern Italy (not covered by Alfani) and the 1600–01 famine reported by Alfani was a relatively minor event, seemingly affecting only Emilia and southern Lombardy. The only relevant discrepancy lies with the 1693–95 event, reported by Alfani only. 1693 is in fact mentioned as a crisis year by Del Panta and Livi Bacci, but without any indication of its causes (save for Milan, where they connect it to smallpox: Del Panta and Livi Bacci [1977]: 418), though more recent literature clearly connects it to dearth or famine (Bellettini 1987: 68–71; Alfani 2010: 40). Some discussion is also needed for the famine of the late 1620s and early 1630s. In fact, this period is much more closely associated with the plague epidemic (the worst since the Black Death in the fourteenth century) that started in late 1629, covered almost all of the North in 1630, and in 1630–31 also affected Tuscany. Given the magnitude of this epidemic, and considering that plague cannot be considered a famine-induced disease (Alfani 2013a: 45–6), it would be very difficult to distinguish the specific impact of famine, which was at most a minor contributor to the overall mortality. Consequently, in the following analyses the occurrence of famine will be tested for 1628–29 only and particular care will be used in interpreting the results.

2 Famines in Central and Northern Italy

Earlier chronologies (Table 2.1) provide us with a number of possible famines whose actual severity we can test, at the local and supra-local levels, by applying the method described in the previous section. In order to do this, we used the largest existing database of Italian time series of burials, first used by Alfani for his study of sixteenth-century general demographic trends and of seventeenth-century plagues (Alfani 2013a; 2013b) and further expanded to cover the whole of the eighteenth century and beyond, and to include part of central Italy (Tuscany and Umbria). This database, which is by far the largest of its kind existing for Italy, comprising 210 distinct communities, allowed us to shape our study of famines chronology as a natural experiment: are the hypotheses found in the literature about the occurrence of famine-related mortality crises in specific years confirmed by our database and methodology, and what can we say

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Table 2.2 *Identifying the main famines in central-northern Italy: communities experiencing 'crisis-level' rises in burials*

	Percentage with crisis	No crisis (n.)	50–99% rise in burials (n.)	100–299% rise in burials (n.)	300% at least rise in burials (n.)
1569–72	50.0	5	5	0	0
1586–87	24.0	19	6	0	0
1590–93	78.8	7	9	7	10
1600–01	30.6	34	7	8	0
1607–08	39.8	56	16	20	1
1618–22	52.5	56	26	33	3
1628–29	47.7	69	23	32	8
1648–49	72.5	44	33	64	19
1678–79	53.8	86	55	41	4
1693–95	65.4	64	60	55	6
1708–09	45.6	87	44	28	1
1716	21.4	121	28	5	0
1724	12.2	137	17	2	0
1764–67	42.1	88	36	24	4
1801	32.5	56	19	6	2

Notes: While the North is covered systematically, of the central Italian regions only Tuscany and Umbria are included in the database.
Source: Database Alfani.

about the intensity and the territorial coverage of specific events? (Note that, in their original study, Del Panta and Livi Bacci used only 15 time series of deaths, of which just eight were for the North, three for Tuscany and four for the rest of Italy, all – with one exception – referring to cities.)

Due to the extreme scarcity of information about burials preceding the second half of the sixteenth century, the analysis covers the period from the 1569–72 famine only. The number of communities available at different periods is highly variable, from just 10 at the time of the 1569–72 crisis up to a maximum of 186 for the 1678–79 crisis. Table 2.2 summarises the results obtained for central-northern Italy as a whole, distinguishing between 'small' crises (with an increase of deaths of 50 to 99 per cent over the normal), 'medium' ones (increases in the 100 to 299 per cent range) and 'major' crises (with a fourfold increase in the level of deaths and more). For the reasons discussed above, crisis year 1672 has not been included in the analysis. Additionally, crisis years 1764 and 1767 have been merged and the whole period 1764–67 has been tested, as, according to Finzi (1986: 356), 1766, which followed a long period of drought, might have been the year when the crisis peaked.

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The data presented in Table 2.2 indicate quite clearly that the most severe crisis was that of 1590–93, confirming that this was 'the worst famine ever faced by Northern Italy during the Late Medieval or Early Modern period' (Alfani 2011: 19). In fact, not only did almost 80 per cent of all the communities considered show the signs of a crisis, but in 10 cases (30.3 per cent) the crisis can be labelled a 'major' one, with deaths increasing by 300 percent or more over the normal (in a single year: also notice that this was a particularly long-lasting famine). The second worst episodes seem to be that of 1648–49, when 72.5 per cent of all the communities considered experienced a mortality crisis and, what is more, in 11.9 per cent of them (19) it was large, and 1693–95, when a crisis is found in over 65 per cent of the communities (71.8 per cent if we exclude the mildly affected central regions from the calculations). One could wonder, though, if this outcome is the result of a sampling bias – for example, in the case of a localised crisis in regions over-represented in the database. To keep this potential issue in check, as well as to give a picture of the territorial coverage of each famine, in Table 2.3 the occurrence of crises has been measured on a region-per-region basis (an exception being made for the 1569–72 episode, due to the small sample size).

The data presented in Table 2.3 confirm the conclusion reached about the particular severity of the famines of 1590–93, 1648–49 and 1693–95. In the case of the first two, the only regions where fewer than half the communities were affected was Piedmont; for the 1590–93 famines at least, this can be explained with the relative ease with which alpine and pre-alpine communities weathered the crisis (Alfani 2007; 2013a). However, this apparent advantage of north-western Italy seems to disappear from the second half of the seventeenth century, an interesting finding that calls into question the issue of causality (see the discussion in the following sections). From the 1670s the situation was usually inverted, with north-eastern Italy showing greater resilience to famine compared to the north-western part of the peninsula. In 1707–08, though, Veneto seems to be affected a bit more severely than Piedmont (though much less than Lombardy). This particular crisis, which is associated with the famous 'Great Winter' that caused widespread famine across Europe, is striking for appearing to be much less momentous than is held by a sizeable literature (Salmelli 1986: 27–41; Bellettini 1987: 97), with just one community out of 160 experiencing a great crisis. Admittedly, however, this was possibly the worst famine occurring in the eighteenth (and nineteenth) century – though the 1764–67 crisis seems to have been of a comparable scale, as also demonstrated by the analysis of changes in food prices in Section 3, and in fact a marked

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Table 2.3 *Proportion of communities affected by a crisis (rise in burials of at least 50 per cent)*

	Piedmont	Liguria	Lombardy	Emilia-Romagna	Veneto	Tuscany
1586–87	0.0	33.3	37.5	16.7		
1590–93	40.0	**100.0**	**80.0**	**100.0**		
1600–01	9.1	22.2	44.4	**50.0**		
1607–08	33.3	46.2	36.4	42.1	36.4	41.7
1618–22	25.0	**61.1**	20.0	**62.1**	**46.2**	**74.1**
1628–29	47.4	25.0	**56.3**	48.3	**80.0**	44.8
1648–49	38.1	**76.0**	**71.4**	**74.4**	**56.3**	**94.7**
1678–79	**71.8**	**69.6**	**50.0**	45.5	33.3	46.5
1693–95	**73.0**	**61.9**	**77.3**	**78.3**	**56.3**	44.2
1708–09	42.9	**66.7**	30.0	37.8	**50.0**	**55.8**
1716	12.5	16.7	35.0	16.2	18.8	27.9
1724	15.2	14.3	0.0	18.4	12.5	9.5
1764–67	26.5	**66.7**	11.1	**52.6**	6.3	**69.8**
1801	46.7		0.0	44.8		11.8

Notes: Regions with at least half the communities affected are in bold. Piedmont includes Aosta Valley, Veneto includes Trentino Alto Adige and Friuli Venezia Giulia, and Tuscany includes Umbria.
Source: Database Alfani.

difference between the pre- and post-1700 famines can be noticed and needs to be underlined.

We could recur to regional-level data to elaborate a kind of index of the overall severity of the famines affecting central-northern Italy, which also takes into account territorial coverage. If we single out the episodes in which at least 50 per cent of the communities of at least three regions experienced a crisis, we discover that the three worst famines all fall into this category, plus what was possibly the fourth worst, in 1678–79, as well as the 1618–22, 1708–09 and 1764–67 events (the period 1628–29 is borderline, though we can presume that at least in some communities – in Piedmont, Lombardy and Veneto in particular – the mortality crisis that occurred in 1629 was due to an early outbreak of the 1629–31 plague). Instead, considering territorial coverage, total intensity expressed as the overall proportion of communities affected, and the occurrence of great (local) crises, the 1586–87, 1600–01, 1607–08, 1716, 1724 and 1801 famines appear to be less important events. These findings improve significantly our knowledge of the Italian famines, first because they allow a better appreciation of the importance of the late

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seventeenth-century events and especially that of 1693–95, overlooked in much of the literature. Second, they put into the right perspective the eighteenth-century famines (1801 included), which on the contrary had been magnified by a specialist literature influenced by accounts of 'European' crises, such as that on the Great Frost of 1708–09 (Lamb 1982; Le Roy Ladurie 1983; Lachiver 1991). Third, our analysis of the occurrence per region and over time of famines has much to offer for a proper discussion of causality (see Section 4).

As already mentioned, it is not possible to recur systematically to time series of burials to check for the occurrence of famines preceding that of 1569–72. However, the fact that Alfani's chronology for the late sixteenth and seventeenth centuries has been largely confirmed by the extensive tests we conducted suggests that, by and large, it should be valid also for the late fifteenth and early sixteenth centuries. Furthermore, Alfani tested the relative importance of sixteenth-century famines by making use of a source much more widely available, for this period, than burial records: the parish books of baptisms, made mandatory for all Catholic parishes by the Council of Trent in 1563, but often available for much earlier (it is a well-known fact that baptisms/births are very responsive to situations of acute scarcity of food: Cattini 1983; Bellettini 1987; Alfani 2007). Regarding the second half of the sixteenth century, such sources confirm the pre-eminent place occupied by the famine of 1590–93 (Alfani 2013a: 56–61). More interestingly, regarding the earlier period, of the eight crises recorded in Table 2.1 that occurred between 1500 and 1560, two can be singled out as the most severe: 1502–05 (a period when plague was also ravaging Italy: Alfani 2013a: 88–9) and 1527–29, 'a general crisis in northern Italy, involving the central and eastern part of the Po Valley, …and sparing only western Piedmont and Liguria' (Alfani 2013a: 56). To these we should probably add the 1539–40 crisis, when high increases in wheat prices were common throughout Italy (see Section 3).

For earlier periods not even baptismal records are available. Consequently, we can rely solely on the 'expert' method, looking for concordance in partial and local chronologies. In Table 2.4 we charted the longest and most complete we could find, overall covering the whole period from about 1250 up until 1470 (when Alfani's chronology in Table 2.1 begins). The chronologies cover the cities of Pistoia (Herlihy 1967) and Florence (Pinto 1978; de La Roncière 2011) in Tuscany as well as Lombardy (Savy 2011). The more general account provided by Corradi (1973 [1865–94]), based on chronicles and other documentation, is also included.

The chronologies we collected suggest that in late medieval Italy, as elsewhere in Europe (Campbell 2010a), the period characterised by the

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Table 2.4 *Compared chronologies of medieval famines, c. 1250–1470*

Pistoia	Florence	Lombardy	Corradi
			1256–58
			1271–72
	1275–77	1276–77	1276–77
	1285–86	1286	1286
	1302–03		1302–03
1313		1311–12	
1328–29	1328–30		1326–30
1339–40	1339–41		1339–40
1346–47	1346–47		1346–47
	1352		1352–53
	1369–70		1368
1375	1375		1374–75
			1384–85
1389			
1393			
1410			1410–12
			1458

Notes: The chronology published for Lombardy covers the period 1270–1330 and includes events of different kinds; we considered only those for which human mortality was reported. The chronology published for Pistoia covers the period 1313–1458 and includes both years of 'scarcity' and of 'famine'; only the latter have been included in the table. Regarding the information provided by Corradi, although he also refers to local famines we included in the list only those episodes that involved the whole of Italy or at least a large area. For Florence, we reconstructed a chronology combining the works of Pinto and de La Roncière, which together cover the period 1250–1375.
Sources: Herlihy (1967: 105) for Pistoia; Pinto (1978; 2012) and de La Roncière (2011) for Florence; Savy (2011) for Lombardy; Corradi (1973 [1865–94]) for other areas.

worst and most frequent famines was that immediately preceding the Black Death. In the first half of the fourteenth century we find four major famines: 1302–03; 1328–30; 1339–40; and 1346–47. As well as the earlier famine of 1275–77, these seem to have been large-scale events, covering the whole of Italy or at least most of it, as also confirmed by the most updated general account of the Italian population of the late Middle Ages currently available (Pinto 1996: 45–6). Instead, Italy seems to have been largely spared by the so-called Great European Famine of 1315–22, triggered by unfavourable weather and further exacerbated by the concomitant cattle panzootic of 1316–25 (Campbell 2010a). It has been hypothesised that, during this crisis, starving England was provisioned with Italian grain, though the evidence for this is limited and caution

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is needed (Jordan 1996: 173–4). However, the advantage of the Italian Peninsula was short-lived, as '[t]he famine of 1328–30 had for Italy the same significance as the one which has been given, for Europe, to the great crisis of 1315–17' (Pinto 1996: 46, our translation). This episode contends with the 1346–47 one for the position as the worst famine of the fourteenth century. Triggered by intense rain since autumn 1345, the 1346–47 famine covered a large European area, particularly the south of the continent. In Florence, for example, city authorities distributed 94,000 certificates allowing the collection of bread from public bakeries: one certificate per head, in a city of 85,000 to 90,000 inhabitants, indicating that the whole of the urban population, as well as many rural dwellers, had to recur to public help (Pinto 2012: 70). To a degree, however, the actual demographic impact of this famine pales compared to the human losses caused by the Black Death, which entered Italy in late 1347 and by the end of the following year had spread to almost all of the peninsula, causing the loss of between 30 and 60 per cent of the entire Italian population (Del Panta 1980; Benedictow 2004; Alfani and Melegaro 2010). In the century or so following the Black Death, famines become relatively rare, and they seem to have been of inferior magnitude compared to those of the early fourteenth century as well as to those of the early modern period – though it should also be noticed that these episodes are clearly under-researched compared to the preceding and following ones.

3 Famines in the Kingdom of Naples and in the Papal States (Latium)

For the southern regions, which during the late medieval and early modern periods were all comprised in the Kingdom of Naples, as well as for part of central Italy and in particular for the region of Latium, which constituted the heart of the Papal States, very few time series of burials of good quality are available, so that we cannot proceed in the same way as for the North and the Tuscan–Umbrian area (in the following, we will indicate these two parts of Italy as Centre-South and Centre-North, respectively). Consequently, we rely on qualitative information and on the specialised literature to construct a famine chronology comparable to that described in the earlier section. An analysis of time series of food prices (of wheat in particular) will be used as an additional tool to both check whether the main events affected the whole of the peninsula, and to deepen our general account of Italian famines. Finally, some information will be provided about the Italian food-provisioning authorities.

Regarding the Middle Ages, the discussion in the earlier section also applies to the Centre-South, and, in particular, the main famines we

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listed for the pre-Black-Death period involved the whole of the peninsula. For Latium, we found confirmation of this in Palermo's (1990; 1997) works on the city of Rome and the *annona*. Consequently, here we will not develop the matter further, focusing instead on the early modern period. A first aspect to be mentioned is that the literature on famines and food security in central-southern Italian regions relies upon even more disparate sources than those available for the Centre-North. Additionally, the two states that covered almost the whole of this large area were characterised by a complex and fragmented territory.

In the Papal States, fragmentation made each province a separate and distinct entity, with its own customs obstructing inter-state trade, its own currency and systems of measurements, which considerably complicated exchanges (Caravale and Caracciolo 1978: 525–36). The Kingdom of Naples was less fragmented, though composed of two separate kingdoms – Naples and Sicily – that, while politically united under the same sovereign, had distinct and deeply different administrations. Particularly relevant for our analysis is the fact that the southern Italian area was divided into many markets, often isolated one from the other due to the lack of adequate land routes – and, consequently, it was a frequent occurrence that food prices differed sharply on different markets of the Kingdom. However, the existence of regional markets and prices did not prevent, during the main famines, grain prices from increasing sharply even in the theoretically most favoured areas. In fact, in these areas hoarding activities, usually very profitable, counteracted the (local) benefices deriving from the existence of natural barriers and the ensuing territorial fragmentation.

In the Centre-South, the first severe famines of the sixteenth century are closely related to warfare, as they occurred during the Italian Wars (1494–1559), triggered by the decision of the French king, Charles VIII, to try and enforce his claim on the Kingdom of Naples. As in the Centre-North, the main crises occurred in 1502–05 and 1527–29 (Alfani 2013a). In the second period famine was exacerbated in Naples by the siege suffered in 1528 at the hands of the French troops led by Lautrec, and in Rome by the sack inflicted by the Landsknechts in 1527 (Alfani and Rizzo 2013: 19–20; Alfani 2013a: 55). Also during the second half of the sixteenth century, as well as in the seventeenth, it seems that the chronology of the main events matches quite closely that already described for the Centre-North. We have some confirmation of this in the dynamics followed by grain prices, as well as – for the seventeenth century – in Del Panta and Livi Bacci's (1977) early work, which included an analysis of time series of burials in three of the main cities of the Kingdom of Naples: Bari, Palermo and Naples itself. Although, for the South, the

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picture provided by Del Panta and Livi Bacci is very sketchy, the worst famine of the late sixteenth and seventeenth centuries is clearly the 1590–93 one. In this period the only other crisis involving all three cities is the 1648–49 one, while in other instances Palermo (the capital city of the island of Sicily, a traditionally wheat-exporting area) seems to have been spared the consequences of severe hunger; this was particularly the case for the 1607–08 and 1618–22 crises. Of the main events identified for the Centre-North, the 1678–79 and 1693–95 famines did not affect any of the three cities; however, both Bari and Palermo (as well as Catania and the rest of Sicily: Fazio 1993; Bulgarelli Lukacs 2009) were affected by the previously mentioned 1672 famine, which possibly involved only the southernmost areas of the Italian Peninsula.

For the eighteenth century a much more sizeable literature is available, and, in particular, the works by Strangio (1998; 1999), Revel (1972; 1975; 1982) and Reinhardt (1991) on the Papal States and by Alifano (1996), Coniglio (1940), Orlandi (1996) and Malanima (2013) for the Kingdom of Naples. In the Papal States, on the grounds of the records of the *annona* and other sources, the main crises (which were essentially due to a scarcity of wheat) occurred in 1708, 1721, 1728, 1743–45, 1748–49, 1764–67 (the worst of all), 1779–80 and 1797 (Strangio 1998). Revel proposes a slightly different chronology of wheat production crises: 1719, 1724, 1744, 1747 and, finally, the whole period 1769–1779 – the 'grand cycle de crise' (Revel 1982: 229). To clarify the matter, Figure 2.1 represents the prices paid by the Roman *annona* to provision wheat, as well as the reference prices that it fixed yearly (usually in September).

Of course, not all these peaks in grain prices are indicative of a real famine (see below for a discussion of the shortcomings of grain prices series as famine markers in the pre-industrial period). 1764–67 and 1779–80 seem to have particular significance, however, as they were characterised both by sharp increases in the prices of wheat and by the activities of the Monte Nuovo Abbondanza delle Comunità, a temporary institution that was aimed at easing, at the local level, the situations of greater scarcity (Strangio 1999; 2013). For the Kingdom of Naples, according to the available literature, food crises occurred in 1723, 1759–64, 1764–66, 1780–81, 1790–91 and 1802. Of these, 1764 seems to have been the most severe (Malanima 2013: 346), as in the Papal States. For the kingdom, significant food crises have been reported for the nineteenth century, too (particularly in 1817, 1854 and 1858: Malanima [2013]: 348).

As can be seen, there are significant analogies between the partial chronologies proposed for Latium/Papal States and for the Kingdom of Naples, as well as for the Centre-North. In order to better distinguish between secondary events and major crises (proper 'famines' covering

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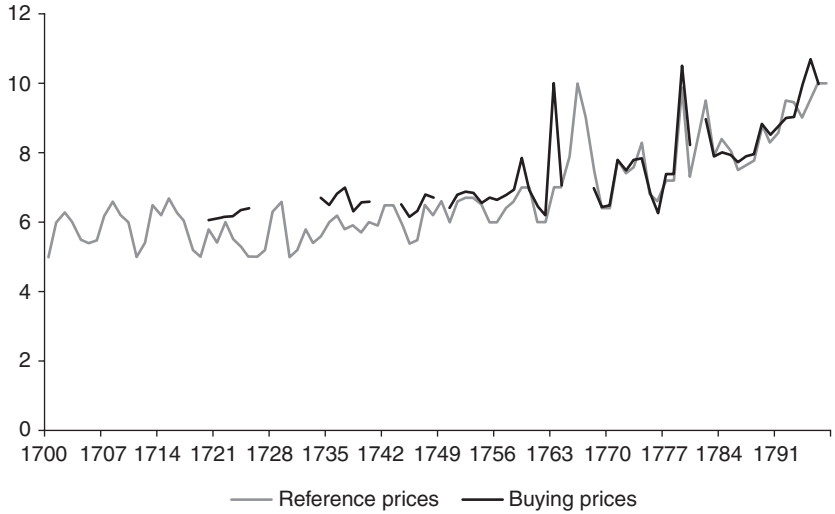


Figure 2.1 Yearly wheat prices of the *annona* (in Roman *scudi*): average buying prices and reference prices (September) compared, 1700–1797

Sources: Archivio di Stato di Roma (ASR), *Presidenza Annona e Grascia* bb. 2212–2249; ASR, Bandi, b. 460; Nicolai (1803: 155); Revel (1972: 255).

a large part of Italy), in Table 2.5 we recur to some of the best and longest time series of wheat prices available, also trying to ensure territorial representativeness across the North, Centre and South of the Italian Peninsula. All prices are average yearly prices on urban markets, and, overall, our time series cover three centuries (about 1500 to 1800). To identify the occurrence of a crisis, we took the peak yearly price found in a crisis period and compared it to 'normal' prices, defined as the seven-term average, max and min values excluded, covering from the eighth to the second year preceding the first crisis year (a procedure similar to that applied earlier to time series of burials). We tested all the main crisis years identified in the earlier section for the Centre-North, plus 1672, 1743–45 and 1779–81. In Table 2.5, cities are arranged along a North–South axis. As an additional check, for the most complete time series we also analysed the deviations of wheat prices from the Hodrick–Prescott smoothed trend (log price, with a smoothing parameter of 6.25). The results are shown in Figure 2.2.

Time series of prices largely confirm the list of major events identified in our systematic analysis of the Centre-North, and allow us to infer that they have much wider general significance. On the contrary, 1672

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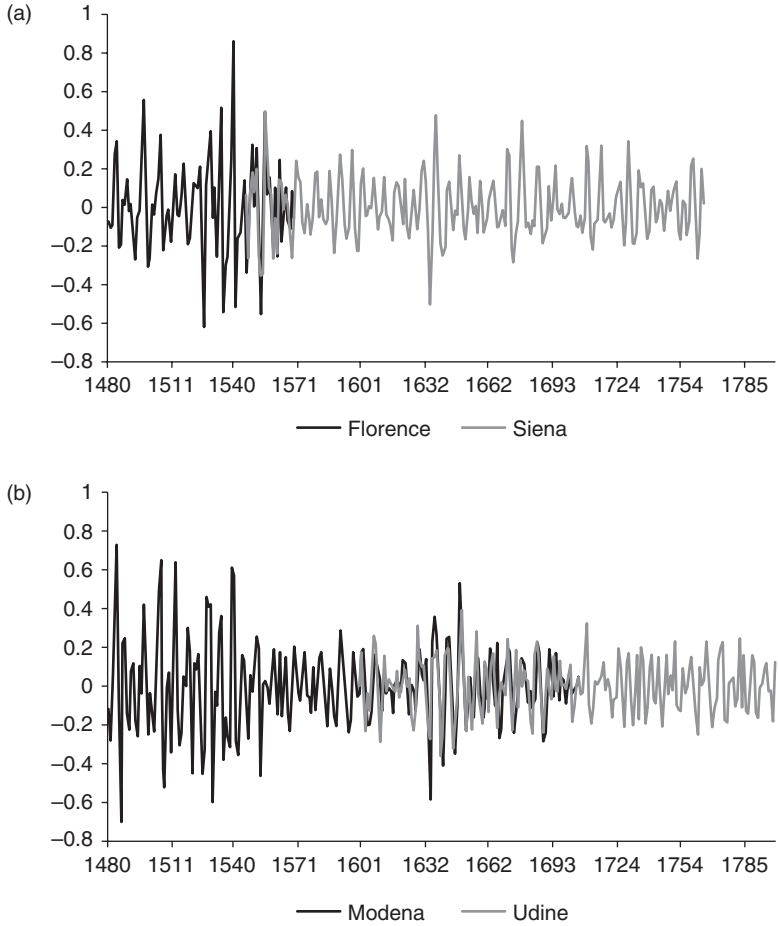


Figure 2.2 Deviations from normal in wheat prices (deviations from Hodrick–Prescott trend of log values), 1480–1799

(a)

Central Italy

(b)

Northern Italy

does not seem to be a year of widespread famine (confirming our earlier hypothesis that the crisis was confined to the southernmost part of Italy), and the same is true for the extra dates tested for the eighteenth century. If we focus on those events that were associated with a doubling

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Table 2.5 *Price increases during potential famines years over normal levels*

	North				Centre						South		
Years	Udine	Bassano	Pavia	Modena	Pesaro	Florence	Sansepolcro	Siena	Rome (buying p.)	Rome (fixed p.)	Naples	Bari	Catania
1502–05			62.1	**163.5**		58.5							
1527–29	**175.2**		**221.0**	**301.1**		**271.4**	**202.9**						
1539–40	77.4		**119.6**	71.5		66.5	**167.7**					**122.4**	38.7
1590–93	82.8	**114.0**	**120.6**	**175.9**		96.8	**123.7**	90.9			59.5	**114.5**	59.3
1618–22	31.0	54.4	32.1	55.4		21.4	92.6	67.8			31.0	46.2	13.1
1648–49	**143.6**	**167.0**	**118.8**	**222.9**				**118.8**			**240.7**	**242.2**	
1672	–2.4	11.2	**9.8**	**27.7**				38.9					
1678–79	27.3	31.0	63.3	39.5				**118.8**			3.2		
1693–95	56.3	71.8		94.1				79.3			37.5		
1708–09	34.0	34.6			20.3			68.7		16.2	61.3		
1743–45	9.5	–26.7			39.1				–1.7	9.8	10.5		
1764–67	60.0	43.2			64.3			63.2	63.3	56.3	63.0		
1779–81	–7.9	–3.8			6.6				40.8	34.5	–1.1		

Note: Increases of 100 per cent or more are in bold.
Sources: Databases Alfani and Strangio.

at least of the average yearly price of wheat (in bold in Table 2.5), we find widespread sharp increases in wheat prices in 1527–29, 1539–40, 1590–93 and 1648–49. Obviously, yearly averages hide much wider intra-year oscillations; for example, the fourfold increase in wheat prices found in Modena during the 1527–29 famine, the almost threefold increase found in the same city in 1590–93 and the increase by three and a half times found in Naples and Bari in 1648–49 should be considered indicative of a really extreme condition of perturbation of grain markets and of entirely compromised access to food. Time series of prices also confirm that eighteenth-century famines were on a much smaller scale compared to those of the preceding two centuries. The largest increases are recorded for the 1764–67 crisis, which, from this perspective, seems to have been worse than the 1708–09 one. Interestingly, on the basis of wheat prices alone, the 1693–95 famine would have to be defined as a medium-sized event. In addition, deviations from the trend suggest that events were more extreme in the sixteenth century and in the first half of the seventeenth compared to later periods. However, it must be noted that time series of prices present some limitations as markers of famine intensity.

First of all, given our definition of famine, food prices are quite an indirect source for considering the final outcome of a crisis: excess mortality. From this point of view, the burial records used for the Centre-North are obviously much better sources. Price increases, in fact, though positively correlated with mortality increases, are a perfect reflection of neither mortality nor – and more importantly – the actual availability of food, as, 'in the case of extreme famines…, food prices fail to fully reflect the gravity of the situation. At a certain point during the crisis, the price paid for wheat and other cereals on the urban markets is no longer an index of the real possibility of finding food in the cities. To put it bluntly, there is a price, but there is no wheat. For the authorities, then, the problem becomes that of finding resources which are to be rationed and distributed to the starving population, mostly freely (practically nobody can afford to pay, and anyway there is no food being sold on the market)' (Alfani 2011: 28). In the Italian case, then, the significance of food prices can be understood only by also taking into account the activities of the public food-provisioning authorities.

Since antiquity, feeding the large cities of the Mediterranean required the establishment of institutions responsible for provisioning. The city of Rome, capital of the Roman Empire, was a pioneer in this, as its needs were far too large to be met solely with the produce of the surrounding countryside. However, over time similar problems led to similar solutions, as recent comparative studies have clearly shown (Marin and Virlouvet

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2003). In Italy, from the Middle Ages urban provisioning institutions looked to the surrounding area, the *contado*, as the first, and usually the main, source of provisions. Asserting their right to a primacy based on ancient medieval traditions, the *annone* aimed to establish a monopoly over 'their' rural areas and to control as far as possible the actual flows of all basic foodstuffs (not only wheat and other grain) entering and leaving the territory (Guenzi 1995; Strangio 1998). The *annone* also published the *calmiere*, a list of reference prices calculated periodically on the basis of prices actually determined, on the urban market, by the meeting of supply and demand. Famine years were the exception, as in those instances the *calmiere* prices had to take into account the resources the poorest had at their disposal (De Maddalena 1949). Another typical function of the food-provisioning institutions was the control and management (direct and indirect) of primary food supplies (Corritore 2007; Alfani 2013a: 73–5).

The development of this complex system of food-provisioning institutions is closely tied to famine chronology, as we know that often the roots of the late medieval and early modern *annone* go back to the great famines of the early fourteenth century. By the fifteenth century the main characteristics of the system were already established; this is, for example, the case for Rome (Palermo 1990: 163). However, it was during the sixteenth century that the control traditionally exercised by the cities on the countryside evolved into more complex forms of organisation, with the final aim of monitoring (and, if need be, regulating) food availability throughout the territory of the state. This happened, for example, in Florence, Milan, Rome (Delumeau 1957–59; Pult Quaglia 1982; Falchi, Pasturi and Sinisi 1995; Palermo and Strangio 1997; Parziale 2009). Even though in many places the new trends were fully developed only in the seventeenth century, often it was the great famine of 1590–93 that provided the necessary stimulus to complete the process of reform, as was the case in Bologna (Guenzi 1978). Although the *annone* were, generally speaking, the main institutions charged with ensuring food security, in times of severe food shortages the intervention of the public could involve other important parts of the administration, up to the highest levels – sovereigns or ruling bodies (Alfani 2015). In this regard, one aspect that needs to be underlined is that the authorities had to demonstrate that they were active and vigilant in order to maintain order and avoid revolts. It is a well-known fact that during famines the crime rate and the general propensity for violence increased dramatically (and city authorities were well aware of this), but as a rule the cause of bread riots was not only hunger – which was, basically, considered a fact of life, like the unreliability of harvests (Guenzi 1995: 292) – but also the suspicion

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of injustice. If it was thought that bread was available but that somebody was hoarding or removing it from the market, or that what bread was available was being unfairly distributed to the advantage of the privileged classes, and that the provisioning institutions and the authorities were not trying to prevent such abuses, then people rebelled in search of justice as well as of food (Thompson 1971; Kaplan 1976; Guenzi 1979; 1995). We have many examples of such revolts during the main famines, as in 1590–93 (Alfani 2013a: 75) and 1648–49 (Guenzi 1979). Violence was not limited to the cities, however, as armed bands roamed the countryside looking for food, further worsening the situation of rural communities, where, during the most severe crises, the traditional ties of solidarity risked collapsing entirely (Alfani 2011; 2013a: 65–70). A more detailed analysis of the functioning and the development across Italy of the food-provisioning authorities during the sixteenth and early seventeenth centuries is provided elsewhere (Alfani 2013a: 70–5).

4 Explaining Famines: Demographic, Environmental, Agrarian and Institutional Factors

Earlier, we reconstructed a reliable famine chronology covering the whole period from about 1250 to the nineteenth century. Now the time has come to provide a synthetic, overall comment about the factors explaining the occurrence of famines – a question notoriously difficult to answer, not least considering how different episodes seem to call for different explanations. However, if we focus on the worst famines of the whole period – i.e. those that seemingly affected all or almost all of Italy, producing a significant increase in mortality – we find that most of them happened at times of high population pressure on the available resources. In Figure 2.3, the current estimates of the population in the Centre-North are represented (by and large, the trend for the whole of Italy would not differ much). As can be seen in the figure, twice during the period (in the early fourteenth century as well as in the late sixteenth/early seventeenth centuries) Italy's population rebounded from a level that corresponds to the maximum carrying capacity, which the peninsula had proved unable to overcome since Roman times (Alfani 2013c). Only during the seventeenth century, and for the first time in history, was that level (corresponding to about 14 to 15 million people in the whole of Italy: Lo Cascio and Malanima 2005) finally exceeded.

The main famines of the late thirteenth century and the fourteenth (1275–77, 1302–03, 1328–30, 1339–40 and 1346–47) happened in a period of acute pressure on the available resources. The same is true for the worst famine of the early modern period, in 1590–93, as well

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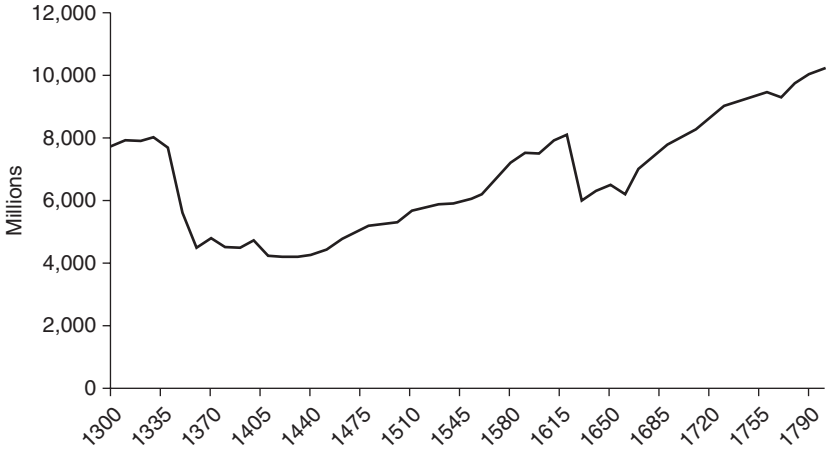


Figure 2.3 Italian population of the Centre-North, 1300–1800

Sources: Database kindly provided by Paolo Malanima, with small corrections around years 1630 and 1656–57 to take into account recent revisions of plague mortality in the seventeenth century (Alfani 2013b).

as for the 1618–22 and 1628–29 ones. Three other general famines occurred when the recovery from the great plagues of the seventeenth century (in 1630 in the Centre-North, minus Liguria; in 1656–57 in the Centre-South, plus Liguria) was completed or on its way to completion: 1678–79, 1693–95 and 1708–09. High pressure on food resources is not sufficient per se to explain the occurrence of a famine, however. No matter how fragile an equilibrium is, something is needed to break it. As a rule in the aforementioned cases, the crisis was triggered by crop failures – usually, back-to-back failures covering two years or more – caused by bad weather, particularly of the kind unfavourable to wheat. For the northern European famines of the early fourteenth century (though the argument can be generalised to the southern European ones), it has been suggested that several episodes of intense cold, as well as incessant rain in the spring, have to be associated with changes in the trend of average temperatures: from cooling to warming (Campbell 2010a; 2010b). Something similar has been suggested for the late sixteenth century and early seventeenth, when the so-called Little Ice Age reached its bottom level (Alfani 2010). It should be underlined that the key factor here was not so much exceptionally low temperatures (as wheat can survive intense cold fairly well) but heavy rain during the spring: exactly the kind of climatic instability that in Italy is empirically associated with an inversion of the temperature trend, from cooling to warming or vice versa (Camuffo

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and Enzi 1992). This was surely the case, for example, with the great 1590–93 famine (Alfani 2011; 2013a: 56–7). It seems, however, that from the second half of the seventeenth century, when a phase of global warming started, another kind of meteorological anomaly – drought – was ever more frequently reported as the 'cause' of food shortages or famines. This would be particularly the case from the mid-eighteenth century onwards (Alfani 2010: 40–1). Another aspect worth mentioning is that the worst plagues of the whole period, the Black Death of 1347–49 and the 1629–30 epidemic, also seem to have occurred in periods of climatic instability. Concerning the Black Death, Campbell (2010a) suggests that climatic instability may have favoured the spread of the disease (as well as of the panzootic that occurred some decades earlier). If this is the case, possibly malnutrition did not contribute significantly to the spread of the plague – a causative link that is in fact contentious (Alfani 2013a: 45–6) – but bad weather 'caused' both famine and plague, so that we would have an additional reason to consider the periods 1346–49 and 1628–31 as marked by long crises, whose characteristics and nature changed during the crisis itself but are referable as a whole to specific causative factors. This is a complex issue, though, which would be worthy of further research.

Apart from the connections between famine and plague, those between famine and war are also worthy of being mentioned. In fact, the two major famines of the first half of the sixteenth century, in 1502–05 and 1527–29, seem to be closely connected to the Italian Wars, when French, Spanish and Imperial troops, in varying alliances with the Italian states, fought for pre-eminence over the peninsula. As a matter of fact, during this exceptionally long conflict the vicissitudes of war weld together many different episodes: general famines, as in the aforementioned periods; local food crises; and simple situations of dearth. Plague and other infectious diseases further complicate the picture (Alfani 2013a: 56–7). What should be stressed here is the ability of human actions both to 'create' famines locally, as in the case of sieges, even in normal years (though, in the medieval and early modern periods, this ability was exerted upon limited areas only) and, more importantly, to worsen the conditions for the production and distribution of food in years of suboptimal crops, allowing a shortage that could have been manageable in other circumstances to develop into a full famine. For example, the food crisis that occurred in Milan in the years 1799–1801, when wheat prices reached their highest value for the period 1700–1860, was due mainly to grain requisitions in order to feed the French army (Mocarelli 2012).

This point is obviously connected to the role played by food-provisioning institutions such as the *annone*, which we briefly described in

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Section 3. In the late medieval and early modern periods it has been suggested that famines were not simply crop failures but have to be understood also as institutional failures, given that those who were charged with keeping the population (or the urban dwellers at least) sufficiently provisioned failed their objective. Obviously, this issue relates to the general debate about whether public or private institutions were better suited to ensuring adequate provisioning. In addition, in the Italian case, advocates of inefficient distribution – possibly exacerbated by the misguided activities of the *annone* – as the main cause of pre-industrial famines are not lacking (Palermo 1990; Epstein 2001), though those underlying production problems and providing a more positive interpretation of the work of the food-provisioning authorities seem to be prevalent. What is more, a detailed recent analysis of the main early modern famines (Alfani 2015) has made the point that, in the worst instances, and particularly during crises covering vast territories and having a marked 'supranational' character, private actors or 'market forces' had neither the strength nor the incentives to provide solutions to the problem. In 1590–93, for example, the Republic of Genoa managed to be the first Italian state to re-establish a situation of acceptable provisioning (even reaping a nice profit from reselling foodstuffs to other states), but it could do so only thanks to the intervention of the highest ruling body (the senate) side by side with the *Magistrato dell'Abbondanza*, the local food-provisioning authority. These public institutions provided the financial means, the diplomatic and political clout and even the ships to open entirely new trade routes to import grain from the Baltic – and, in doing so, contributed to changing for good the structure of grain commerce in the wider European area (Alfani 2015). The relative successes of the Italian food-provisioning authorities in the sixteenth century should not hide the fact, however, that, in general, the literature seems to be quite less favourable to the actions of the *annone* in the eighteenth century (Reinhardt 1991; Strangio 1999).

Of all the main famines we have identified, only the 1648–49 and 1764–67 ones have not been already mentioned and need some additional comment. For 1648–49, we know from Del Panta and Livi Bacci (1977) that famine mortality was further increased by widespread epidemics of typhus, which is a typically famine-related disease. Regarding 1764–67, we know that drought affected the Italian countryside for several years in succession (Finzi 1986: 356). If we consider that, in this period, population density in the Italian Peninsula was steadily increasing to unprecedented levels, we might imagine that population pressure was again playing an important role. However, for the eighteenth century, the occurrence of (relatively mild, in our long-run perspective) famines

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is less interesting than the fact that the ancient maximum carrying capacity had been exceeded, and that, as will be remembered from Section 2, north-eastern Italy seems to be the front-runner of improved food security. It is probable, in fact, that both processes have the same cause: the spread of maize, whose key importance in allowing Italy to escape the 'Malthusian' barrier that had constrained it for so long has recently been underlined (Alfani 2010; 2013c). More generally, maize seems to have contributed very significantly to improving Italian food security, not only because it led to a substantial increase in calories per hectare but also because it prospered in meteorological conditions adverse to wheat, so that Italian farmers, especially in the Centre-North, started to cultivate this new crop imported from the Americas as a kind of insurance against the failure of wheat (Finzi 2009). This process had been under way since the early seventeenth century (even earlier in some areas: Alfani 2012), and can be described as a 'Boserupian'[1] process of agrarian innovation put in place by a population feeling a growing pressure on the resources available (Alfani 2007; 2013a; 2013c). In north-eastern Italy (Veneto) the spread of maize was particularly intense, a fact that is usually considered a sign of the progressive decline into relative poverty and underdevelopment of the area, also due to the spread of the pellagra disease, a form of severe avitaminosis that can result from a diet excessively reliant upon maize. (Something similar happened in the State of Milan, where maize production more than doubled in the second half of the eighteenth century and where the first complaints about the presence of pellagra date to the first half of the same century: Mocarelli [2015: 42–45].) Apparently, pellagra was the price that Veneto had to pay for the exceptionally good food security it acquired during the eighteenth century. This is a paradoxical conclusion, which allows us to stress one last time the complexity of events such as famines, whose boundaries are never as clear-cut as first imagined, and whose causes and implications can be deeply different from one instance to another – though, for exactly the same reasons, these dramatic events represent one of the most fascinating topics for historical research.

[1] According to Boserup (1965; 1981), demographic pressure itself promotes technological micro-innovations to traditional practices in agriculture and permits a slow increase in total population by starting a chain reaction.

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